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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering (CSE)

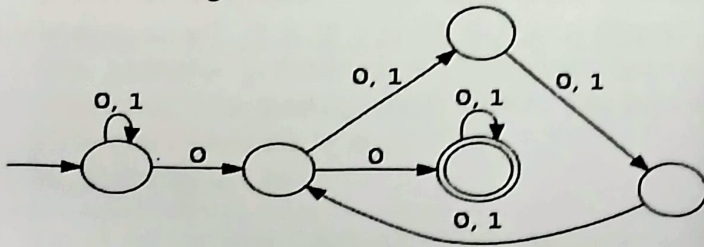
End-Term Examination May - 2025

Subject Name: **Formal Language & Automata Theory**
MM: 60

Subject Code: **CSE-225**
Time Allotted: **180min**

NOTE: 1. All questions are compulsory.
2. Mark answer number clearly.

1(a) Consider the following NFA.



(i) Informally (English language statement) but clearly explain the language accepted by above NFA. (1)

(ii) Write simple regular expression corresponding to above NFA. (2)

(iii) Find the minimum state DFA that accepts the same language. (2)

(b) (i) Design a DFA that will accept all the strings of $\{0, 1\}$ in which every substring of length 3 contains at most two 0's. (3)

(ii) Design an NFA that will accept the set of strings of $\{a, b, c\}$ such that the last symbol in the input string also appears previously in the string. (2)

a) (i) Consider the following context free grammar over alphabet $\{a, b\}$ (3)

G1: $S \rightarrow aSb | \epsilon$

G2: $S \rightarrow Sb | A$

$A \rightarrow aA | \epsilon$

The language generated by Grammar G1 and G2 are $L(G1)$ and $L(G2)$ respectively. Find the language L generated by $L(G1) \cup L(G2)$.

Also find and justify the type of language L from the set = {Regular, CFL but not Regular, CSL but not CFL}.

(ii) Prove that: "DCFL Languages are closed under complementation". (2)

b) Find CFG for the following CFL. (5)

(i) $L = \{a^m b^n c^p d^q \mid n+p = m+q \text{ and } m, n, p, q \geq 0\}$

(ii) $L = \{n_a(w) \neq n_b(w) \mid w \in \{a, b\}^*\}$

Q. 3(a) PDA for the language $L = \{n_a(w) = n_b(w) + n_c(w) \mid w \in \{a, b, c\}^*\}$

(b) Let the following PDA $M = (\{q_0, q_1, q_2\}, \{0, 1, 2\}, \{0, 1, z\}, \delta, q_0, \{q_2\})$ with the following transitions:

$\delta(q_0, 0, z) = \{(q_0, 0z)\},$
 $\delta(q_0, 0, 1) = \{(q_0, 01)\},$
 $\delta(q_0, 1, 1) = \{(q_0, 11)\},$
 $\delta(q_0, 2, 1) = \{(q_1, 1)\},$
 $\delta(q_1, 1, 0) = \{(q_1, \epsilon)\},$
 $\delta(q_1, \epsilon, z) = \{(q_2, z)\}$

$\delta(q_0, 0, 0) = \{(q_0, 00)\},$
 $\delta(q_0, 1, z) = \{(q_0, 1z)\},$
 $\delta(q_0, 1, 0) = \{(q_0, 10)\},$
 $\delta(q_0, 2, 0) = \{(q_1, 0)\},$
 $\delta(q_1, 0, 1) = \{(q_1, \epsilon)\},$

where z indicate bottom marker of the stack. Find and justify the language accepted by the above PDA.

Q. 4(a) Consider the following CFG

$S \rightarrow SaS \mid SbS \mid c$

Is the above grammar an ambiguous grammar? If Yes then generate all the parse trees corresponding to any string of length 7 ($|w| = 7$).

(b) Construct PDA for the following Context Free Grammar $G(\{S, A, B, C\}, \{a, b, d\}, S, P)$. The set of productions

P of G are as follows:

$S \rightarrow aS / Aa$
 $A \rightarrow Cb / B$
 $B \rightarrow bA / Cd$
 $C \rightarrow Sd / \lambda$

Q. 5(a) Construct Context Sensitive Grammar for the following Languages.

(i) $L = \{a^n b^m c^n d^m \mid n, m > 0\}$

(ii) $L = \{a^n b^n a^n \mid n > 0\}$

(b) Find TM for the language $L = \{a^n b^m c^p d^q \mid n = p \text{ and } m = q \text{ and } m, n, p, q \geq 1\}$

Q. 6(a) Construct PDA to CFG

$\delta(q_0, a, z) = \{(q_0, Az)\},$
 $\delta(q_1, b, A) = \{(q_2, A)\},$

$\delta(q_0, a, A) = \{(q_1, AAA)\},$
 $\delta(q_2, c, A) = \{(q_0, \epsilon)\},$

(b) Describe the Chomsky hierarchy of languages in details.